SWIGGY SALES ANALYSIS

DESCRIPTION:

The Swiggy dataset contains 197,430 records and 10 columns related to restaurants, food items, pricing, customer ratings, and locations across different cities in India. It includes information such as restaurant names, food categories, dish names, food prices, ratings, rating counts, city, state, and order dates. This dataset is used to analyze restaurant distribution, food pricing trends, customer ratings, and category-wise performance using SQL.



PROBLEM STATEMENT:

Swiggy aims to analyze restaurant and food-related data across different cities in India to understand restaurant distribution, food pricing, customer ratings, and food category trends. The objective is to use SQL to identify patterns, evaluate restaurant performance, and generate meaningful business insights that can support better business decisions and improve customer experience.

BUSINESS REQUIREMENTS:

Data Cleaning & Validation

The raw table swiggy_data contains food delivery records across states, cities, restaurants, categories, and dishes.

The first goal is to ensure data quality by performing:

Null Check

Identify missing values in:

- State
- City
- Order_Date
- Restaurant_Name
- Location
- Category
- Dish_Name
- Price_INR
- Rating
- Rating_Count

Blank/Empty String Check:

Detect fields containing blank values that may cause inaccurate analysis.

Duplicate Detection:

Find duplicate rows using grouping on all business-critical columns.

Duplicate Removal:

Use ROW_NUMBER() to delete surplus duplicate rows while retaining one clean copy for each unique order.

Dimensional Modelling (Star Schema):

Dimensional modelling helps organize data in a way that makes analysis simple, consistent, and extremely efficient. Instead of keeping all information mixed in one large table, the Star Schema separates descriptive information into dimensions and keeps all measurable values in a central fact table. This structure reduces data duplication, improves clarity, and makes the entire dataset easier to understand. It also supports faster reporting because queries can access small, focused tables rather than scanning a single bulky dataset. Most analytics and BI tools are designed to work best with star schemas, so this approach ensures smooth dashboard creation, accurate aggregations, and reliable insights. Overall, dimensional modelling provides a clean, scalable, and performance-optimized foundation for any analytical system, regardless of the industry or data source.

To optimize analytics and reporting, build a **Star Schema** with the following dimension tables:

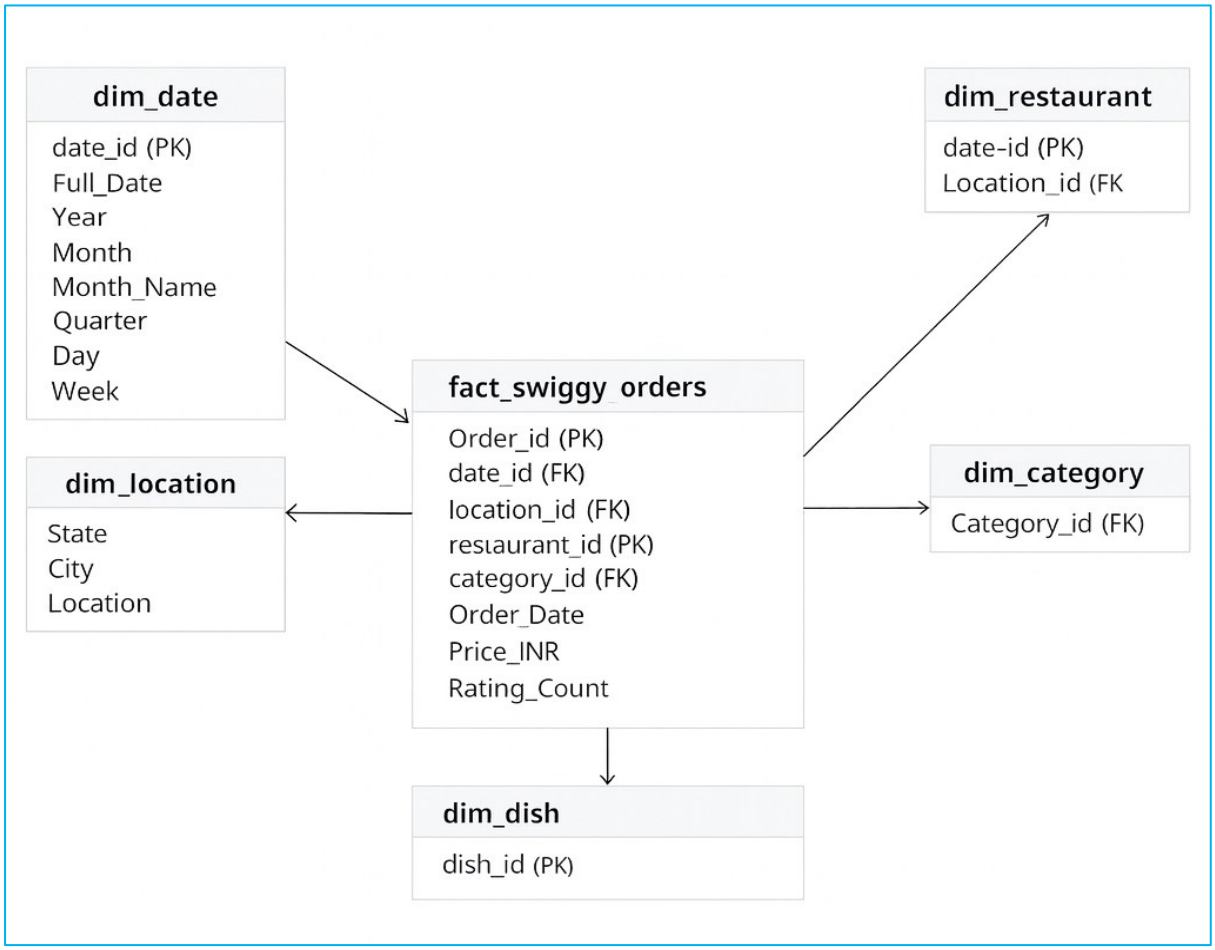
- **dim_date** → Year, Month, Quarter, Week
- **dim_location** → State, City, Location
- **dim_restaurant** → Restaurant_Name
- **dim_category** → Cuisine/Category
- **dim_dish** → Dish_Name

Central fact table:

- **fact_swiggy_orders** → Price_INR, Rating, Rating_Count, foreign keys to all dimensions

Populate each dimension with **distinct** data from the cleaned source and load fact table with all keys resolved.

ERD DIAGRAM: STAR SCHEMA



KPI Development

Once the schema is built, compute core performance indicators:

Basic KPIs

- Total Orders
- Total Revenue (INR Million)
- Average Dish Price
- Average Rating

Deep-Dive Business Analysis

Date-Based Analysis

- Monthly order trends
- Quarterly order trends
- Year-wise growth
- Day-of-week patterns

Location-Based Analysis

- Top 10 cities by order volume
- Revenue contribution by states

Food Performance

- Top 10 restaurants by orders
- Top categories (Indian, Chinese, etc.)
- Most ordered dishes
- Cuisine performance → Orders + Avg Rating

Customer Spending Insights

Buckets of customer spend:

- Under 100
- 100–199
- 200–299
- 300–499
- 500+

With total order distribution across these ranges.

Ratings Analysis

Distribution of dish ratings from 1–5.